

Task 3 Report

Patient Retrieval

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INTRODUCTION

The objective of Task 3 is to evaluate the quality of the learned patient embeddings using a clinically intuitive retrieval paradigm. The underlying hypothesis is straightforward: if the embeddings capture clinically meaningful information, patients that are close in the latent space should also exhibit similar clinical characteristics.

To test this hypothesis, a patient retrieval framework based on cosine similarity was implemented. For each query patient, the nearest neighbours in embedding space were retrieved using a leave-one-out strategy for three neighbourhood sizes ($k = 5, 10, \text{ and } 20$). The clinical concordance between the query patient and the retrieved neighbours was then evaluated using several complementary metrics.

This task provides a complementary perspective to the analyses performed in Tasks 1 and 2. While Task 1 investigated whether clinical variables could be predicted from the embeddings using supervised learning, and Task 2 assessed whether the embeddings naturally organised into clinically coherent groups in an unsupervised setting, Task 3 focuses on the local neighbourhood structure of the latent space and evaluates whether clinically similar patients are located close to one another.

From a clinical perspective, such a retrieval paradigm is particularly relevant because it closely resembles potential real-world applications of representation learning in precision medicine. Identifying patients with similar molecular profiles could support patient stratification, prognosis estimation, treatment response prediction, or cohort enrichment for clinical trials.

Based on the findings of the previous tasks, several hypotheses can be formulated. First, Level 2 (GR-DAE) embeddings are expected to provide the highest retrieval precision for PAM50 molecular subtypes, owing to the beneficial effect of PPI regularisation on subtype-related information observed in Task 1. Second, Level 3 (GS-DAE) embeddings are expected to provide the strongest retrieval performance for survival, reflecting the prognostic information captured by graph propagation through the PPI network. Finally, pathological stage is expected to remain difficult to recover across all representation levels, consistently with the results observed in both Tasks 1 and 2.

The analysis therefore focuses on three clinically relevant variables: PAM50 molecular subtype, survival group, and pathological stage. The following sections describe the retrieval protocol,

present the results obtained for each clinical variable, and compare the performance of the different representation levels.

1 EXPERIMENTAL PROTOCOL

1.1 RETRIEVAL STRATEGY

The retrieval analysis was performed independently on each of the four embedding levels (Levels 0 to 3). For each patient, cosine similarity was computed with all other patients in the embedding space using a leave-one-out strategy, ensuring that a patient could not retrieve itself as a neighbour.

For every query patient, the top-k most similar patients were retrieved for three neighbourhood sizes:

- $k = 5$
- $k = 10$
- $k = 20$

These values were selected to evaluate retrieval performance at different scales, ranging from immediate local neighbourhoods to broader patient communities.

Cosine similarity was chosen because it measures the angular similarity between embedding vectors independently of their magnitude and is widely used for similarity search in representation learning applications.

1.2 CLINICAL CONCORDANCE METRICS

The clinical similarity between each query patient and its retrieved neighbours was evaluated using three complementary metrics corresponding to the main clinical variables investigated throughout this project.

PAM50 retrieval precision

PAM50 retrieval performance was evaluated by measuring the proportion of retrieved neighbours sharing the same molecular subtype as the query patient.

Higher values indicate better retrieval quality, with a value of 1 corresponding to perfect agreement, meaning that all retrieved neighbours belong to the same PAM50 subtype as the query patient.

Values close to the random baseline indicate that the embedding space does not preserve subtype-specific neighbourhoods.

Because PAM50 represents an established molecular classification of breast cancer, this metric constitutes the primary evaluation criterion of Task 3.

Survival retrieval precision

Survival retrieval performance was assessed by computing the proportion of retrieved neighbours belonging to the same survival group as the query patient.

As for PAM50, higher values indicate better neighbourhood consistency, with a value of 1 corresponding to perfect agreement between the query patient and all retrieved neighbours. Values close to the empirical random baseline indicate that the embedding space does not preserve prognostic neighbourhoods beyond what would be expected by chance.

Unlike the supervised classification experiments of Task 1, this metric evaluates local neighbourhood similarity rather than global class separability.

Stage concordance

For the pathological stage, retrieval quality was assessed by measuring the mean absolute difference in stage between the query patient and its retrieved neighbours.

In contrast to the previous metrics, lower values indicate better performance. A value of 0 corresponds to perfect concordance, meaning that all retrieved neighbours belong to the same pathological stage as the query patient, whereas larger values indicate increasing disagreement between neighbouring patients.

Since the pathological stage follows an ordinal progression, measuring the magnitude of disagreement provides a more informative evaluation than simply considering exact matches.

1.3 RANDOM BASELINE

To quantify the practical utility of the embeddings, retrieval performance was compared with a random retrieval baseline.

For PAM50, the expected random precision approximately corresponds to the prevalence of the dominant subtype in the TCGA-BRCA cohort, namely Luminal A, resulting in an expected baseline close to 0.31.

For survival and stage, the corresponding random baselines were estimated empirically by repeatedly sampling random neighbours and computing the associated concordance metrics.

This comparison makes it possible to determine whether the learned representations capture clinically meaningful neighbourhood structures beyond what would be expected by chance.

2 PAM50 RETRIEVAL PERFORMANCE

The retrieval performance for PAM50 molecular subtypes was evaluated using Precision@k for each representation level and each neighbourhood size. The results are summarised in the table below and visualised in Figure 1.

Level	K = 5	K = 10	K = 20
Level 0	0.228	0.227	0.227
Level 1	0.245	0.240	0.236
Level 2	0.345	0.328	0.310
Level 3	0.251	0.246	0.240

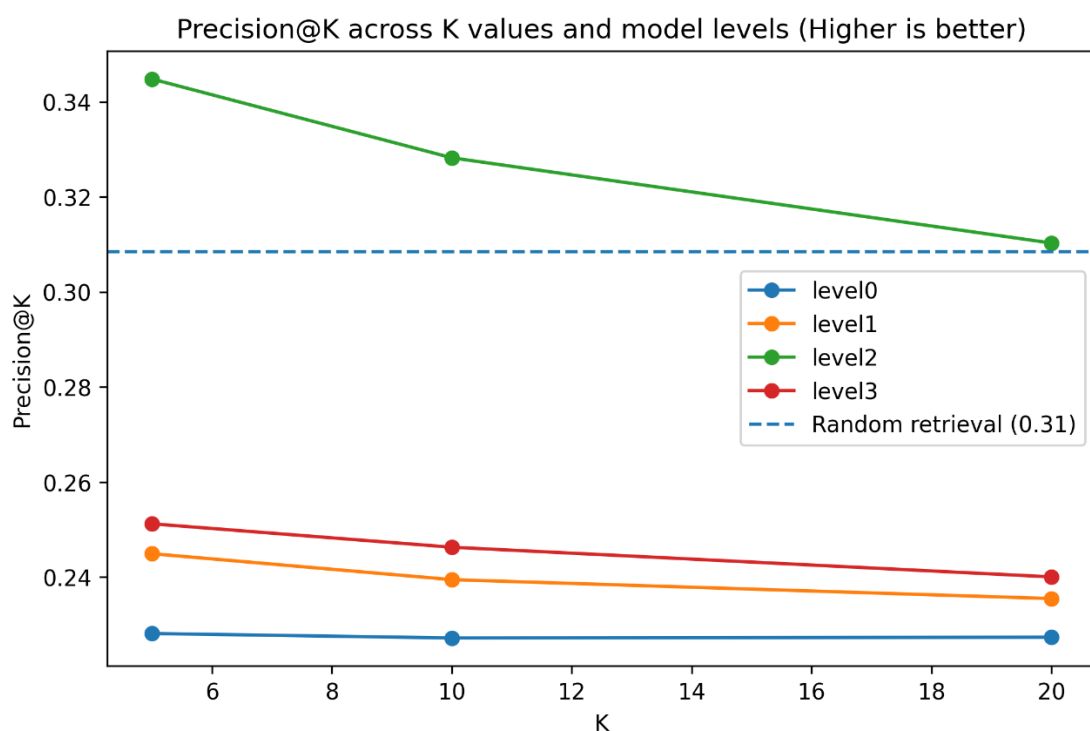


Figure 1 : Precision@k curves for PAM50 across representation levels.

2.1 INTERPRETATION

Level 2 (GR-DAE) is the only representation level that consistently outperforms the random baseline across all neighbourhood sizes, reaching a Precision@5 of 0.345. This finding indicates that the graph-regularised denoising autoencoder successfully organises patients with similar molecular subtypes into local neighbourhoods of the embedding space.

In contrast, Levels 0 (AE), 1 (DAE), and 3 (GS-DAE) remain below the random baseline for all values of k , suggesting that their neighbourhood structures do not preserve PAM50 subtype information sufficiently to support clinically meaningful retrieval.

These findings are highly consistent with the supervised probing results obtained in Task 1, where Level 2 (GR-DAE) embeddings also achieved the highest classification performance for PAM50 prediction. Together, these results suggest that PPI-based graph regularisation improves not only global class separability but also local neighbourhood coherence for molecular subtype information.

A second observation is that Precision@ k decreases slightly as the neighbourhood size increases for all representation levels. This behaviour indicates that the closest neighbours tend to be more clinically similar than more distant neighbours, which is an expected and desirable property for retrieval-based applications.

Finally, the relatively poor performance of Level 3 (GS-DAE) provides additional evidence supporting the oversmoothing hypothesis formulated in Task 2. While graph propagation through the PPI network improves prognostic information, it appears to attenuate the fine-grained molecular signatures required to preserve PAM50 subtype-specific neighbourhoods.

3 SURVIVAL RETRIEVAL PERFORMANCE

The retrieval performance for survival groups was evaluated using Precision@k for each representation level and each neighbourhood size. The results are summarised in the table below and visualised in Figure 2.

Level	K = 5	K = 10	K = 20
Level 0	0.787	0.779	0.776
Level 1	0.764	0.768	0.772
Level 2	0.775	0.775	0.776
Level 3	0.820	0.820	0.822

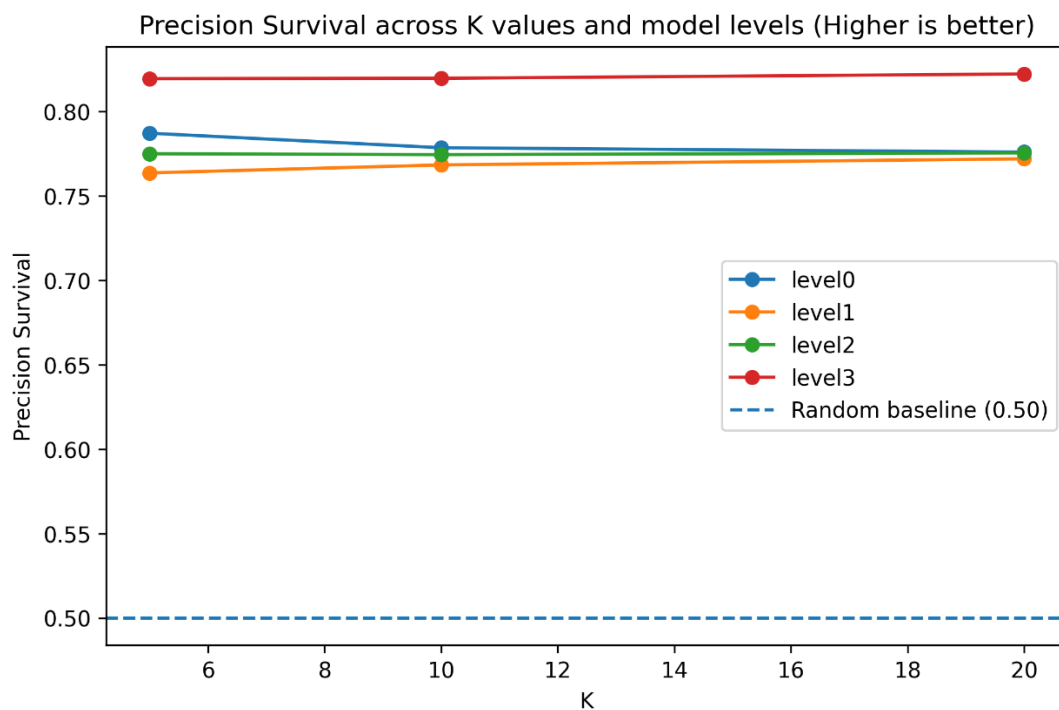


Figure 2 : Precision@k curves for survival across representation levels.

3.1 INTERPRETATION

All representation levels substantially outperform the random baseline of 0.50, with Precision@5 values ranging from 0.764 for Level 1 (DAE) to 0.820 for Level 3 (GS-DAE). These results indicate that patients with similar survival outcomes tend to occupy neighbouring regions of the embedding space, regardless of the representation level.

Level 3 (GS-DAE) consistently achieves the highest retrieval performance across all neighbourhood sizes, reaching a Precision@5 of 0.820 and a Precision@20 of 0.822. This finding further supports the hypothesis that graph propagation through the PPI network captures biologically meaningful prognostic information and organises patients according to survival-related patterns.

A particularly interesting observation emerges when comparing these results with the supervised classification experiments of Task 1. In the supervised setting, Levels 0 (AE) to 2 (GR-DAE) exhibited near-random performance for survival prediction ($F1 \approx 0.50$ and $AUC \approx 0.50$), suggesting the absence of exploitable prognostic signal for global classification. In contrast, the retrieval analysis reveals strong survival concordance within local neighbourhoods for all representation levels.

This apparent contradiction suggests that survival information is already present in the lower-level embeddings, but is encoded in a local and distributed manner that cannot be easily recovered by global linear decision boundaries. Similar patients tend to cluster together in the latent space, even though the overall geometry does not permit accurate separation of survival groups at the population level.

These findings highlight the complementary nature of supervised probing and similarity-based retrieval for evaluating representation quality. While supervised classifiers assess global class separability, retrieval-based approaches evaluate local neighbourhood consistency and may therefore reveal clinically meaningful information that remains hidden to conventional predictive models.

Finally, Precision@k remains remarkably stable across neighbourhood sizes for Level 3 (GS-DAE), suggesting that the prognostic organisation of the embedding space extends beyond the immediate neighbours and remains preserved over larger patient neighbourhoods.

4 STAGE CONCORDANCE

The retrieval performance for the pathological stage was evaluated using stage concordance, defined as the mean absolute difference in stage between each query patient and their retrieved neighbours. Lower values indicate better performance, with 0 corresponding to perfect concordance.

The results for the original multi-class stage variable are presented first, followed by a binary version of the task designed to investigate whether simplifying the stage hierarchy could improve retrieval performance.

4.1 MULTI-CLASS STAGE CONCORDANCE

The stage concordance results for the original five-class stage variable (0, I, II, III, IV) are reported in the table below and visualised in Figure 3.

Level	K = 5	K = 10	K = 20
Level0_stage	0.696	0.688	0.688
Level1_stage	0.683	0.688	0.696
Level2_stage	0.702	0.694	0.701
Level3_stage	0.658	0.665	0.668

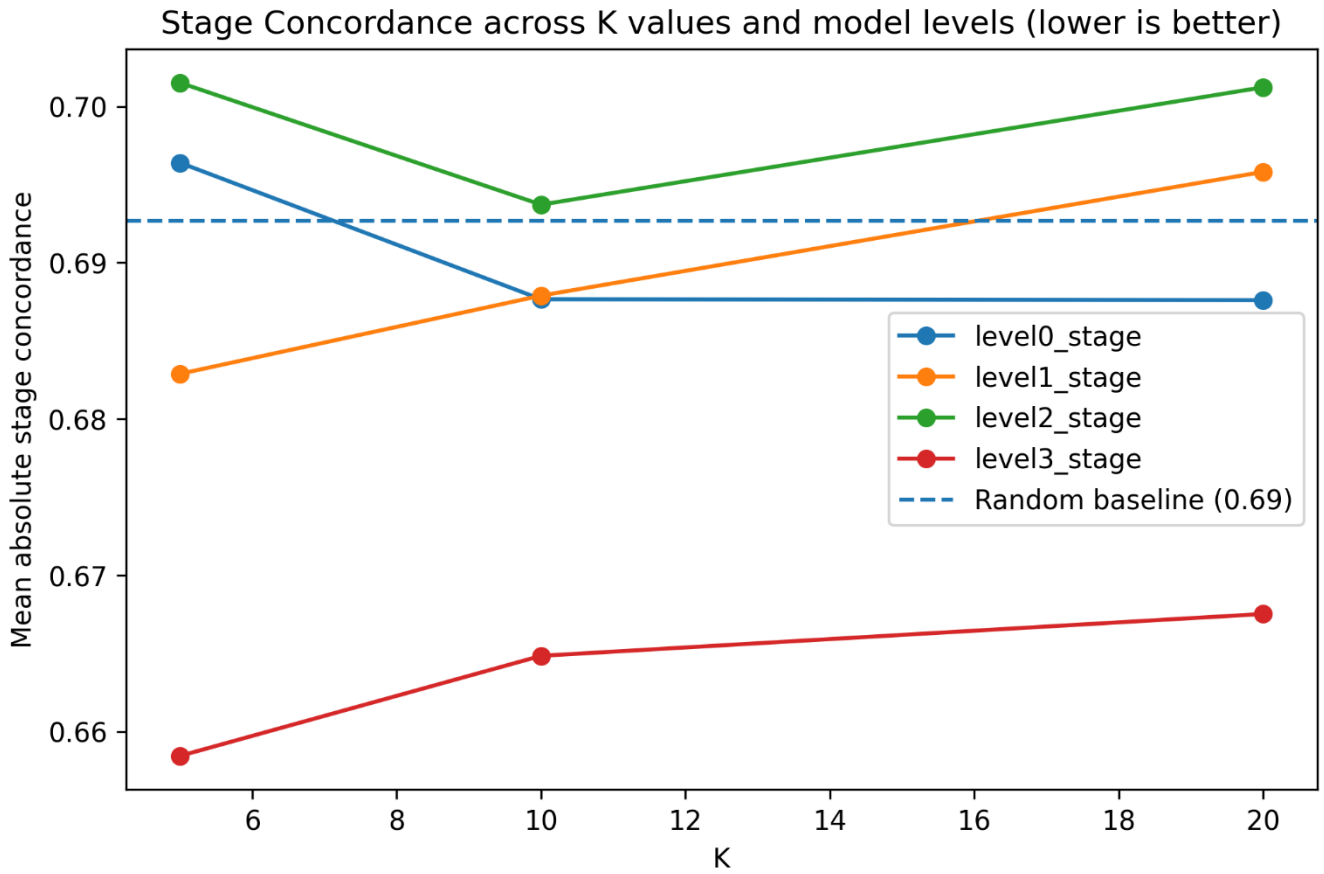


Figure 3 : Stage concordance curves across representation levels.

All representation levels exhibit stage concordance values close to the random baseline (~0.69), with no level consistently outperforming the others. Level 3_stage shows slightly lower values (0.658 at $k=5$), suggesting a marginal improvement, but the difference remains modest.

These results are consistent with the observations from Tasks 1 and 2, where stage prediction and clustering proved equally challenging.

4.2 BINARY STAGE RETRIEVAL

To further investigate whether stage information could be recovered under a simplified setting, a binary version of the stage variable was constructed by grouping stages 0, I, and II as “early” (0) and stages III and IV as “advanced” (1). Precision@ k was then computed for this binary task.

The results are summarised in the table below and visualised in Figure 4.

Level	K = 5	K = 10	K = 20
Level0_stage	0.656	0.654	0.653
Level1_stage	0.652	0.650	0.649
Level2_stage	0.663	0.657	0.650
Level3_stage	0.664	0.662	0.661

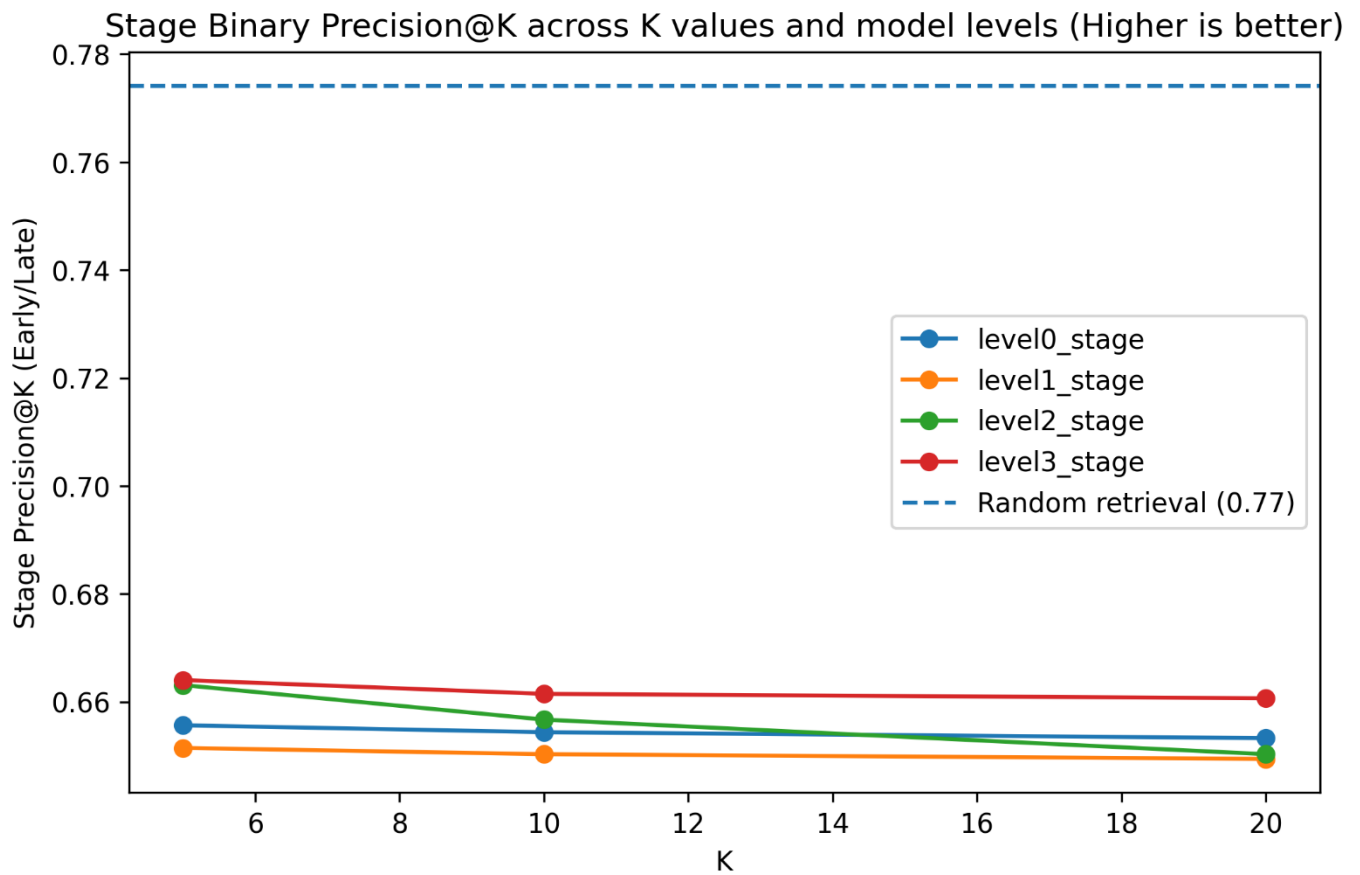


Figure 4 : Precision@k curves for binary stage (early vs advanced).

The random baseline for this binary task is approximately 0.77, reflecting the predominance of early-stage patients in the cohort. All representation levels remained below this baseline, indicating that collapsing the stage categories into a simpler binary task was not sufficient to recover clinically meaningful neighbourhood structure.

Level 3_stage achieves the highest Precision@5 (0.664), but this improvement remains insufficient to outperform random retrieval, confirming that the embeddings contain limited stage-related signal regardless of the task formulation.

4.3 INTERPRETATION

The persistent difficulty in recovering stage information across classification (Task 1), clustering (Task 2), and retrieval (Task 3) suggests that the pathological stage is only weakly represented in the learned embeddings.

A first possible explanation is the severe class imbalance present in the TCGA-BRCA cohort, particularly for Stage 0 and Stage IV patients, which may hinder the learning and evaluation of stage-related patterns. To investigate this possibility, a simplified binary formulation distinguishing early-stage (0–II) from advanced-stage (III–IV) disease was introduced.

However, the binary retrieval experiment did not substantially improve performance. Despite collapsing the original classes into a simpler binary formulation, all representation levels remained below the random baseline, indicating that class imbalance alone cannot explain the observed results.

These findings therefore suggest that the limited performance arises from a combination of class imbalance and weak stage-related biological signals within the transcriptomic embeddings themselves. Unlike molecular subtype or survival information, the pathological stage may depend more strongly on anatomical and clinical factors that are not fully captured by gene expression profiles alone.

Future work could investigate whether oversampling strategies (e.g., SMOTE) improve retrieval performance, or whether integrating additional clinical modalities could enhance the representation of disease progression within the embedding space.

5 STATISTICAL VALIDATION

To ensure that the observed performance differences reflect true biological signals rather than sampling variation, a rigorous statistical validation framework was implemented. For all retrieval configurations, 95% confidence intervals for the Precision@k metrics ($k \in \{5, 10, 20\}$) were calculated using bootstrap resampling (1000 resamples). To rigorously assess whether the representations outperform the random baseline and whether specific representation levels surpass others, paired permutation tests were performed on the per-patient retrieval scores (with significance declared at $p < 0.05$).

The results confirm distinct statistical patterns across the clinical endpoints:

- **PAM50 Subtype Retrieval:** Level 2 emerges as the only representation level to significantly outperform the random baseline (0.308) for PAM50 retrieval at smaller neighbourhood. For instance, when evaluating the immediate neighbourhood at $k = 5$, we are 95% confident that the true population Precision@5 for Level 2 lies between 0.328 and 0.362, significantly surpassing the random baseline of 0.308 ($p < 0.001$). This significant advantage is maintained at $k = 10$, where we are 95% confident the true Precision@10 falls between 0.315 and 0.342 ($p = 0.003$). However, as the neighbourhood expands to $k = 20$, the performance of Level 2 yields a 95% confidence interval ranging from 0.300 to 0.321, which no longer reaches statistical significance compared to the baseline ($p = 0.357$). This indicates that subtype-related local coherence diminishes beyond the immediate neighbourhood. By contrast, Levels 0, 1, and 3 consistently fail to surpass the random baseline across all values of k , with non-significant p -values. Paired permutation tests further confirm that Level 2 is significantly superior to all other levels at $k = 5$ and $k = 10$ ($p < 0.001$ in all pairwise comparisons).
- **Survival Retrieval:** All four representation levels significantly outperform the random baseline (0.5) for survival retrieval across all values of k ($p < 0.001$ in all cases), confirming that survival-related biological structure is encoded across the full model hierarchy. Level 3 achieves the highest performance consistently. At the strictest neighbourhood size ($k = 5$), we are 95% confident that the true population Precision@5 for Level 3 lies between 0.801 and 0.837. This robust performance remains stable as the neighbourhood expands: at $k = 10$, the 95% confidence interval spans from 0.803 to 0.836, and at $k = 20$, it ranges from 0.806 to 0.838. Paired permutation tests confirm that Level 3 is significantly superior to all other levels at all neighbourhood sizes ($p < 0.001$), with performance differences of +0.032 to +0.056 over the next best level. Notably, the advantage of Level 3 remains stable as k

increases, indicating that the GNN encoder captures survival-related biological structure more robustly and consistently than simpler architectures.

Level 2 embeddings provide clinically significant PAM50 neighbourhood structure at small to moderate neighbourhood sizes, while Level 3 embeddings provide significantly superior and robust survival neighbourhood structure across all k values. The two levels encode complementary clinical information, with statistical significance confirmed through both bootstrap confidence intervals and paired permutation tests (1000 permutations).

CONCLUSION

This study evaluated the quality of patient embeddings using a clinically intuitive retrieval paradigm. The underlying hypothesis was that if the embeddings capture clinically meaningful information, patients that are close in the latent space should also exhibit similar clinical characteristics.

The results demonstrate that the different representation levels encode complementary biological information. Level 2 (GR-DAE) consistently achieves the best retrieval performance for PAM50 molecular subtypes, with a Precision@5 of 0.345, outperforming the random baseline of 0.31. This suggests that the graph-regularised denoising autoencoder organises local neighbourhoods according to molecular subtype information, in line with the supervised classification results from Task 1 and the significant clinical associations observed in Task 2.

In contrast, Level 3 (GS-DAE) provides the strongest retrieval performance for survival risk, reaching a Precision@5 of 0.820, substantially above the random baseline of 0.50. This indicates that the GNN encoder captures prognostic information effectively, and that patients with similar survival outcomes are located close to one another in the Level 3 (GS-DAE) embedding space. This observation is consistent with the superior survival prediction performance observed in Task 1 and with the strong clinical associations identified in Task 2.

Interestingly, all representation levels substantially outperform the random baseline for survival retrieval, including Levels 0 to 2, despite these same representations exhibiting near-random survival classification performance in Task 1. This contrast suggests that survival-related information is present within the local neighbourhood structure of the embeddings even when it cannot be recovered through global linear decision boundaries. These findings highlight the complementary nature of supervised probing and similarity-based retrieval for evaluating representation quality.

Stage remains the most challenging clinical target across all representation levels. Even after simplifying the task to a binary formulation (early versus advanced disease), retrieval performance remained below the random baseline, suggesting that stage-related information is only weakly encoded in the embeddings. This observation is consistent with the difficulty observed in both supervised classification (Task 1) and unsupervised clustering (Task 2) and likely reflects a combination of severe class imbalance and limited stage-related biological signal within transcriptomic data.

Across all tasks, classification (Task 1), clustering (Task 2), and retrieval (Task 3), Level 2 (GR-DAE) emerges as the most informative representation for molecular subtype information (PAM50),

whereas Level 3 (GS-DAE) is consistently the most informative for prognostic information (survival). No single representation level dominates across all clinical objectives, highlighting the complementary nature of graph regularisation and graph propagation.

These findings indicate that clinically relevant information is encoded within the embeddings, but that it is distributed differently across representation levels. Graph regularisation in Level 2 (GR-DAE) appears to preserve local molecular signatures that facilitate subtype discrimination, whereas graph propagation in Level 3 (GS-DAE) promotes the emergence of broader prognostic patterns that are particularly relevant for patient outcomes.

From a clinical perspective, neighbourhood-based representations could support future applications in patient stratification, prognosis estimation, and case-based decision support. Identifying patients with similar molecular profiles and clinical trajectories may help guide treatment selection, improve risk assessment, or facilitate cohort enrichment in clinical studies.

Overall, the retrieval paradigm provides a valuable complementary validation of embedding quality by evaluating clinical similarity at the individual patient level rather than global predictive performance alone. More generally, the results suggest that graph-based representation learning captures clinically meaningful patterns that extend beyond predefined clinical taxonomies and may provide complementary information for future precision oncology applications. Future work could explore the application of these embeddings to additional clinical endpoints or cancer types, as well as the biological interpretation of individual latent dimensions and patient neighbourhoods.